Alzheimer’s Disease

Alzheimer’s disease is a neurodegenerative brain disorder that causes progressive deterioration in memory, thinking, and the ability to maintain daily activities over time. Alzheimer’s disease is the most common form of dementia and accounts for approximately 60–70% of dementia cases. Dementia, on the other hand, is not a specific disease itself, but rather a condition characterized by impairment in cognitive functions such as memory, thinking, communication, and the ability to perform daily activities. Therefore, not all cases of dementia are due to Alzheimer’s disease; however, Alzheimer’s disease is the most common cause of dementia.

One of the most notable characteristics of Alzheimer’s disease is that its symptoms generally develop slowly and progressively. Memory problems are among the first symptoms noticed in many individuals. In particular, individuals may forget recently experienced events, repeatedly ask the same questions, or lose belongings and place them in unusual locations. In addition, a person may begin to experience difficulties with everyday tasks that they previously performed with ease, such as driving, preparing meals, or paying bills. Difficulty finding the right words, impairments in reasoning and decision-making abilities, visuospatial difficulties, and becoming lost even in familiar places may also occur during the early stages of the disease. As the disease progresses, behavioral changes, restlessness, and agitation may develop; in advanced stages, individuals may eventually require assistance even with basic daily activities such as eating and walking.

The development of Alzheimer’s disease cannot be attributed to a single cause. Age is the most important known risk factor, and the disease becomes more common particularly in older age. However, genetic characteristics, lifestyle, and certain environmental or metabolic factors may also influence the risk of developing the disease. Diabetes, cerebrovascular disease, traumatic brain injury, stress, poor diet, and low levels of physical activity have been associated with an increased risk of Alzheimer’s disease. This indicates that the disease is not solely related to genetic factors or aging, but rather may arise through a multifactorial process involving the interaction of different factors.

From a genetic perspective, the vast majority of Alzheimer’s disease cases do not result from a single causative gene. Instead, multiple genetic characteristics may influence disease risk in combination with an individual’s lifestyle and environmental factors. One of the best-known genes in this context is the APOE gene. The APOE ε4 variant increases the risk of Alzheimer’s disease, whereas the ε2 variant is thought to have a potentially protective effect. However, it is important to note that not everyone who carries an APOE ε4 allele will develop Alzheimer’s disease. In contrast, rare mutations in the APP, PSEN1, and PSEN2 genes can cause certain inherited forms of early-onset Alzheimer’s disease, particularly those that begin before the age of 65. Therefore, the statement that “Alzheimer's disease is inherited within families” is not applicable to every patient; although genetic susceptibility is important, most cases of the disease are not attributable to a single inherited genetic cause.

So, what does Alzheimer’s disease do inside the brain? The disease initially affects brain regions associated with memory; as the disease progresses, areas involved in language, reasoning, and social behavior are also affected. Among the major neuropathological features of Alzheimer’s disease are the accumulation of amyloid-beta (Aβ) in the brain, the formation of neurofibrillary tangles composed of hyperphosphorylated tau protein within neurons, and the progressive loss of synapses. These changes disrupt the normal functioning of neurons and their communication with one another. Over time, many neurons lose their connections and eventually die.

However, Alzheimer’s disease is not simply a matter of amyloid and tau protein accumulation. Numerous mechanisms are thought to contribute to the development and progression of the disease, including vascular abnormalities, mitochondrial dysfunction, oxidative stress, reduced glucose utilization in the brain, and neuroinflammation. Brain insulin resistance has also been associated with processes such as increased oxidative stress, increased Aβ42 production, and tau protein phosphorylation. In addition, evidence suggests that neuroinflammatory mechanisms and pathways such as the NLRP3 inflammasome may contribute to the disease process. Therefore, Alzheimer's disease is regarded as a complex disorder involving numerous interconnected biological mechanisms.

At present, there is no definitive cure for Alzheimer's disease. Nevertheless, certain medications can be used to control symptoms or to help preserve memory and thinking abilities to some extent. To reduce the risk of developing the disease, maintaining physical activity, following a healthy diet, avoiding smoking and harmful alcohol use, remaining socially and cognitively active, and controlling health indicators such as blood pressure, blood glucose, and cholesterol are considered important.

In conclusion, Alzheimer's disease should not simply be regarded as a disease of forgetfulness. In addition to impairments in memory, it is a progressive brain disorder that can increasingly affect thinking, decision-making, language, behavior, and the ability to perform daily activities, with genetic susceptibility, age, lifestyle, and numerous biological mechanisms contributing to its development. A better understanding of the mechanisms underlying the disease is therefore of great importance for the development of earlier diagnostic approaches and more effective treatment strategies.

References:

1) World Health Organization. (2026, July 3). *Dementia*. https://www.who.int/news-room/fact-sheets/detail/dementia

2) National Institute on Aging. (2024, February). *Understanding Alzheimer's Disease Genes*. Alzheimer's and Related Dementias Education and Referral Center, National Institutes of Health. https://order.nia.nih.gov/publication/understanding-alzheimers-disease-genes

3) National Institute on Aging. (n.d.). *What is Alzheimer's disease?* National Institutes of Health. https://www.nia.nih.gov/health/alzheimers-and-dementia/what-alzheimers-disease

4) Rostagno AA. Pathogenesis of Alzheimer's Disease. Int J Mol Sci. 2022 Dec 21;24(1):107. doi: 10.3390/ijms24010107. PMID: 36613544; PMCID: PMC9820480.

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